

Crash Course in Derivatives*

Here we are going to introduce the operation of the derivative *algebraically* and then later we will go into some, often quite surprising, geometric intuitions.

First we are going to take the derivative of some polynomials: we usually use the notation $\frac{d(\text{function})}{d(\text{variable})}$ or $\text{function}'$ to denote a derivative, the former is from Leibniz and the latter is from Newton (I prefer the former, something I will explain when we get to the geometric interpretation). For functions we will use our normal function notation $f(x)$ = some function of x but be careful when considering expressions with multiple variables. So this means $\frac{d}{dx}p(x) = p(x)'$ for some polynomial $p(\cdot)$ using the variable x .

Suppose you were given the following expressions as true:

1. If $p(x)$ is a constant, $p'(x) = 0$
2. If $p(x) = x$ then $p'(x) = 1$
3. If $p(x) = x^2$ then $p'(x) = 2x$
4. If $p(x) = x^3$ then $p'(x) = 3x^2$
5. If $p(x) = x^6$ then $p'(x) = 6x^5$

What would the following derivatives of the following functions look like:

1. $g(x) = x^5$
2. $k(x) = x^{1001}$
3. $f(x) = x^{-2}$
4. $h(x) = x^{\frac{2}{3}}$
5. $j(x) = x^{\frac{-2}{3}}$

With those in mind, what would the more general rule look like? if $p(x) = x^n$ then

$$p'(x) = ???$$

An interesting property of the derivative, one that comes up a lot in various fields of mathematics is called *linearity*. Linearity for derivatives has three major conditions:

- **Null Preservation:** The derivative of 0 is 0 (and since we already said the derivative of a constant is 0, and 0 is a constant, this checks out!)
- **Sub-additivity:** The derivative of the sum of two functions is equal to the sum of their respective derivatives added together. In symbols:

$$(f(x) + g(x))' = f'(x) + g'(x)$$

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- **Scalar Preservation:** Multiplying a function by a constant and then taking its derivative is equal to first taking its derivative and then multiplying by the same constant. In symbols

$$(c \cdot f(x))' = c \cdot f'(x)$$

Here is an example of our second and third rule combined:

$$\left(2x^3 + \frac{1}{3}x^{-3}\right)' = 6x^2 - x^{-4}$$

Generate 4 more examples to get a feel for how these rules work, these will come in handy when we get to more complex functions:

1. Function: (eg. $r(x) = 2x^{1/3} - 2x^{-1/3}$)

Derivative:

2. Function:

Derivative:

3. Function:

Derivative:

4. Function:

Derivative:

Normally, in a real Analysis class, we would go through all of the properties and rules for derivatives using a more exploratory mindset, meaning we would figure the following rules out using an argument that we each discovered ourselves. For now, however, I am just going to give you the algebraic rules for the *multiplication rule* of derivatives, and then, once we have our feet wet algebraically, we can do some discovery (if not together, then at home with fancy online resources like brilliant.org or khanacademy.org).

Multiplication Rule: The multiplication rule looks as follows (in symbols):

$$(f(x) \cdot g(x))' = f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

Sometimes we might even see this shortened (without all the brackets) to be something like the following, which might make it easier to visualize or remember:

$$(uv)' = u'v + v'u$$

Regardless of how you write it, the first thing one should do when introduced to a new rule is check it: If we write x^5 as a multiplication of two smaller polynomials ($x^2 \cdot x^3$) for example, then we can try out this rule and see if it works:

$$\begin{aligned}(x^5)' &= (x^2 \cdot x^3)' = (x^2)' \cdot x^3 + (x^3)' \cdot x^2 \\ &= 2x^1 \cdot x^3 + 3x^2 \cdot x^2 = 2x^4 + 3x^4 = 5x^4 = (x^5)'\end{aligned}$$

Give the multiplication rule a try for the following:

1. $f(x) = (x^2 + 2x + 1)$, $g(x) = (x + 1)^2$
2. $h(x) = 2x^3 - 2x^{-3}$, $j(x) = \frac{1}{2}x^{1/3}$
3. Can you generate two non-constant functions that when multiplied together equal one of the functions?
 $((f(x)g(x))' = g(x)$
4. what is $(f(x)f(x))'$ equal to for any differentiable $f(x)$? What problems might arise?